

Evidentiary Architecture in Cancer Biomarker Science

A Critical Reading of “Bridging Discovery and Treatment: Cancer Biomarker”

A critical analysis of Liu, Karsidag, Golin & Wu (2025), *Cancers*

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1. Research Background

Precision oncology depends on a working assumption that is easy to state and hard to operationalize: that molecular signals correlating with tumor behavior can be measured reliably enough, and interpreted precisely enough, to change what a clinician does for a given patient. The evidentiary distance between “this molecule correlates with outcome” and “this molecule should determine treatment” is where most of the field's actual difficulty lives, and it is not a distance that a single assay type — genomic, transcriptomic, proteomic — can close on its own. Liu et al.'s review sits inside that gap. Rather than defending one biomarker class, it maps the landscape of what is measurable across sample types (tissue, blood, imaging) and molecular modalities (DNA, RNA, epigenetic, protein, metabolite, microbial), and names where that landscape currently fails to reach clinical practice.

That framing sets the question I read the rest of the paper against: does surveying breadth this wide produce a coherent account of translational readiness, or does it flatten meaningfully different evidence levels into a single, undifferentiated inventory of “promising biomarkers”?

2. Paper Overview

Paper reviewed: “Bridging Discovery and Treatment: Cancer Biomarker” (2025), Hengrui Liu, Ilayda Karsidag, Rebecca Golin & Guangzhen Wu, *Cancers* (MDPI), 17(22), 3720.

Type: Narrative review. Area: Precision oncology, cancer biomarker science, translational research.

The authors' stated organizing structure moves through detection modalities and sample types, clinical applications (early detection, theranostics, monitoring), the biomarker's role in treatment decision-making (target identification, resistance detection, early and real-time intervention), a dedicated section on translational barriers, and a closing discussion of personalized-medicine strategy (tumor heterogeneity, multi-omics integration, cost, AI-driven personalization). It is explicitly a synthesis of existing literature rather than a study reporting new experimental results, and it should be read as such: every biological and clinical claim in the paper is a claim about what cited primary studies report, filtered through the authors' selection and framing, not a new finding this review itself generated.

3. Conceptual Framework

3.1 Framework Overview

The review's classification scheme operates on three semi-independent axes:

Sample type — tissue biopsy, liquid biopsy (ctDNA, CTCs, extracellular vesicles), imaging, and non-molecular clinical parameters.

Detection dimension — local bulk vs. global bulk profiling (a single-site tissue signal vs. a blood-based signal aggregated across the whole tumor burden), and single-cell vs. spatially resolved profiling (cellular resolution without tissue context vs. tissue context without single-cell resolution).

Molecular modality — DNA, RNA, epigenetic, protein, metabolite, and microbial signal.

These axes are not mutually exclusive, and the review does not claim they are — a ctDNA fragmentomic assay is simultaneously a liquid-biopsy sample type, a global-bulk detection dimension, and a DNA-level modality. Where the classification is useful is in making explicit that a biomarker's clinical behavior depends jointly on all three axes, not on molecular class alone. Where it becomes strained is in section 3.4, where a

short discussion of prion-detection assays (RT-QuIC, PMCA) is introduced without being situated on any of the three axes — it reads as an appended example of a detection principle (protein-misfolding amplification) rather than a genuine extension of the paper's own framework.

4. Key Findings

Framed correctly, “findings” for a review of this kind means claims the authors synthesize from cited primary and secondary literature, not results this paper itself generated. That distinction matters throughout, because the review's persuasiveness depends partly on how faithfully it represents the evidentiary strength of what it cites — a point developed further in Critical Analysis.

4.1 Findings Summary

A representative, non-exhaustive sample of what the review reports:

ctDNA-based fragmentomic profiling (fragment-length distribution, end-motif patterns) is presented as improving specificity for minimal residual disease detection relative to mutation-calling alone, with residual ctDNA detectable, per the cited literature, months before radiographic relapse in some cohorts.

MGMT promoter methylation status in glioblastoma is presented as one of the clearest examples in the paper of a biomarker incorporated into treatment decision-making: methylation status is described as informing selection for temozolomide, not merely as predicting who is likely to respond — though the review does not suggest methylation status alone universally determines treatment, and other clinical factors remain part of that decision.

PSMA PET/CT paired with PSMA-targeted radionuclide therapy is used as the paper's central illustration of a theranostic biomarker — one where the same molecular target defines both the diagnostic and the therapeutic intervention.

The ESMO Precision Medicine Working Group recommendations, published in 2022, propose a dual-validation approach (cfDNA screening followed by confirmatory tissue NGS) that reads not as a resolution of the liquid-vs-tissue biopsy disagreement but as a procedural accommodation of it — an acknowledgment that neither modality alone currently meets the reliability bar for standalone clinical decision-making.

Section 5.6 argues that machine-learning-based biomarker discovery is vulnerable to a specific, named failure mode — data leakage arising from premature feature selection, inadvertent use of future timepoints, or hyperparameter tuning informed by test-set performance — and cites the “Model Information Sheets” proposal (Kapoor & Narayanan, 2023) as a partial reporting-standard remedy.

4.2 Section Contributions

The review does not itself organize its examples by evidentiary stage. Reconstructing that placement — Discovery → Analytical Validation → Clinical Validation → Clinical Utility → Regulatory Adoption → Routine Implementation — for a representative subset of the paper's own examples makes the range of evidentiary maturity the review is actually covering explicit:

Biomarker example	Pipeline stage (reconstructed)	Basis for placement
MGMT methylation (glioblastoma)	Routine implementation	Cited as directly informing treatment selection (temozolomide) in current practice

Biomarker example	Pipeline stage (reconstructed)	Basis for placement
EGFR / BRAF companion mutation testing	Regulatory adoption / routine implementation	Approved companion diagnostics tied to specific inhibitor therapies
ctDNA fragmentomics (MRD detection)	Clinical validation, approaching utility	Detects relapse ahead of imaging in cited cohorts; ESMO dual-validation guideline reflects utility not yet fully established for cfDNA alone
Exosomal / EV protein & RNA markers	Analytical / clinical validation	Associations with resistance and relapse reported; no prospective utility trial cited for these specific markers
Microbiome composition (e.g., Akkermansia, ICI response)	Discovery / early clinical validation	Correlative association reported; authors themselves flag contamination artifacts and batch effects as unresolved
AI-driven multi-modal phenotyping	Discovery	Presented as an emerging direction; no clinical validation evidence cited for the integration claim itself
Prion-based amplification assays (RT-QuIC / PMCA)	Conceptual / pre-discovery in oncology	Presented as a detection-principle analogy, not as an oncology biomarker with reported evidence

What this reconstruction makes visible is that the review's prose register does not track this range particularly well. A biomarker at routine implementation and a biomarker still at the discovery stage are frequently described using structurally similar declarative sentences, and a reader relying on tone rather than checking each evidentiary chain would have difficulty telling them apart on the page.

5. Critical Analysis

5.1 Strengths

Multi-axis classification (sample type × detection dimension × molecular modality).

What it does well: makes explicit that two biomarkers of the “same” molecular class can differ substantially in what they can and cannot detect depending on detection dimension. Why it matters: without that explicit framing, biomarker comparisons across studies risk conflating assay class with assay performance. Value added: gives a reader a structured way to ask what tradeoff a specific assay is making, rather than treating biomarker discovery as a single undifferentiated activity.

A dedicated translational-barriers section with a structured summary table.

What it does well: names six distinct failure points (validation and trial design, regulatory/IVD registration, standardization, privacy/ethics, methodological controversy, ML reproducibility) rather than treating “translation is hard” as a single vague caveat. Why it matters: each failure point has a different remedy and timescale, and conflating them makes the barrier appear more uniform, and therefore easier to dismiss with a generic call for further research, than it actually is. Value added: a genuine decomposition of the translational bottleneck rather than a rhetorical acknowledgment of it.

A dedicated subsection on machine-learning reproducibility (section 5.6).

What it does well: imports a specific, mechanistically described failure mode — leakage via premature feature selection, temporal leakage, test-set-informed hyperparameter tuning — into a review whose primary audience is unlikely to be machine-learning specialists. Why it matters: computational biomarker discovery is increasingly central to the field this review surveys, and reported model performance is exactly

the kind of number that is easy to over-trust without this context. Value added: this subsection does real epistemic work that the rest of the review does not consistently apply to itself, discussed further in 5.4.

5.2 Weaknesses

No consistent evidence-level tagging.

What is missing: the review does not consistently distinguish, at the point each claim is made, whether the underlying evidence is preclinical, retrospective, or prospective, or whether it has cleared analytical, clinical, or only correlative validation. Why it matters: a reader moving through the paper linearly encounters MGMT methylation (routine clinical use) and microbiome-immunotherapy correlations (early discovery-stage) in structurally similar declarative sentences. Consequence: the prose does not, on its own, signal which claims are load-bearing for clinical decisions today and which are promissory. What would improve it: even a simple evidence-stage annotation attached to each subsection, of the kind reconstructed in 4.2, would substantially change how a first-time reader calibrates confidence claim by claim.

Limitations are proportionally under-discussed relative to claims.

What is missing: several subsections (metabolomics, epigenetics, microbiomics) append a genuine, nontrivial limitation as a subordinate clause at the end of an otherwise affirmative paragraph rather than giving it independent treatment. The metabolomics subsection is representative — pre-analytical variability and rapid ex vivo degradation of metabolites are described in six words at the very end of a sentence that is otherwise about metabolomics' promise as a therapeutic-vulnerability biomarker. Why it matters: pre-analytical variability is not a minor caveat in metabolomics; it is one of the primary reasons metabolomic biomarkers have struggled to reach cross-cohort reproducibility. Consequence: a reader without prior familiarity with the field could reasonably conclude the limitation is minor because of where and how briefly it appears. What would improve it: limitations of this magnitude warrant separated, explicit discussion proportional to how much they constrain the preceding claim.

The review does not audit its own citation base for independence.

What is missing: a subset of the review's citations — specifically references 34, 36, and 140–142 — are prior first-author publications by the review's own corresponding author. In the resistance-biomarker section this is handled appropriately: the authors cite their own prior bioinformatic analyses specifically as examples of retrospective, in-silico work requiring prospective validation, a self-critical rather than self-promotional use. Elsewhere — the T-cell exhaustion-marker citation supporting a claim about checkpoint-inhibitor resistance — the self-citation functions as ordinary supporting evidence, indistinguishable in the prose from any other citation. Why it matters: this is not evidence of misconduct, and self-citation is normal where a research group has genuinely contributed multiple relevant studies. But a review's persuasive force depends partly on the independence of its evidence base, and the text gives the reader no way to distinguish these two citation functions without checking the reference list directly. Consequence: a reader assessing how much independent literature supports a given claim cannot do so from the prose alone. What would improve it: even a brief acknowledgment of the authors' own prior contributions to the cited literature would let readers calibrate this appropriately.

5.3 Advantages vs. Limitations, at a Glance

Advantage	Corresponding Limitation
Multi-axis classification clarifies detection tradeoffs across biomarker classes	No parallel structure exists for evidentiary stage; a taxonomically well-placed biomarker can still be evidentially immature

Advantage	Corresponding Limitation
Translational-barriers section decomposes a vague problem into six named failure points	Mitigation strategies for each barrier are described at a programmatic level (“early coordination,” “ethical frameworks”) rather than evaluated for feasibility
ML-leakage subsection applies real methodological rigor to computational biomarker discovery	That same rigor is not extended to the AI-driven phenotyping claims made earlier in the paper (3.2, 6.4)
Section 3.5 attempts a genuine comparative synthesis across clinical contexts	This synthesis is not extended to most of the paper's other sections, which remain primarily descriptive

5.4 Evidence and Methodological Scrutiny

Biological, analytical, and clinical validity, applied directly.

The review does not itself apply the biological / analytical / clinical validity / clinical utility distinction explicitly. Locating three of its central examples against it directly is informative about what the paper's claims actually establish.

MGMT promoter methylation in glioblastoma. Biological validity: methylation silences a DNA-repair gene, and the mechanistic link between reduced repair capacity and increased alkylating-agent sensitivity is well established in the literature the review draws on. Analytical validity: the review does not report sensitivity, specificity, or inter-laboratory concordance figures for MGMT methylation assays specifically, though methylation-specific PCR and pyrosequencing platforms for this marker are in wide clinical use, which is indirect evidence of reasonable analytical maturity. Clinical validity: methylation status is associated with treatment response in the cited literature. Clinical utility: this is the strongest case in the paper for utility rather than mere validity — methylation status is described as informing which patients are selected for temozolomide, meaning the biomarker factors into management rather than only risk stratification, although the review does not indicate that methylation status alone should determine that decision independent of other clinical considerations.

ctDNA fragmentomics for minimal residual disease. Biological validity: tumor-derived cfDNA fragments differ systematically in length and end-motif pattern from non-tumor cfDNA, a distinction with a plausible biological basis in differential nuclease processing, though the review does not itself detail the mechanism. Analytical validity: not quantified in the review — no sensitivity, specificity, or limit-of-detection values are reported for fragmentomic assays specifically. Clinical validity: the review cites evidence that residual ctDNA detection precedes radiographic relapse by months, a genuine clinical-validity claim, since it is an association with a clinically meaningful endpoint. Clinical utility: not established in what the review presents — detecting relapse earlier is not automatically equivalent to demonstrating that earlier intervention triggered by that detection improves outcomes, and no interventional trial testing that specific hypothesis is cited for fragmentomic MRD assays.

Microbiome composition and checkpoint-inhibitor response. Biological validity: a plausible mechanism (modulation of systemic inflammation and antigen presentation) is proposed. Analytical validity: the review itself flags contamination artifacts and batch effects in low-biomass microbial sequencing as an active, acknowledged problem — a direct admission of immature analytical validity. Clinical validity: the association is described using correlational language throughout (“correlates with,” “likely by”), not causal or predictive language. Clinical utility: not claimed by the review; adjunctive strategies (dietary intervention, probiotics, fecal microbiota transplantation) are described as being “explored,” which is accurate, explicitly exploratory language.

The pattern across these three examples is informative on its own: the review's rhetorical register does not change very much between the MGMT example, which has traveled the full distance to clinical utility, and the microbiome example, which has not yet cleared analytical validity.

Computational and machine-learning biomarker discovery.

Section 5.6 is, on its own terms, a methodologically careful treatment of a real problem. It correctly identifies that inflated reported performance in ML-based biomarker studies frequently traces back to information leaking from the test set into training, whether through premature feature selection, temporal leakage, or hyperparameter tuning informed by test-set performance. The Model Information Sheets proposal it cites addresses this at the level of reporting standards rather than algorithmic design, which is an appropriately scoped, documentation-based remedy. This is a legitimate application of the question this kind of critique should ask: can reported computational performance be interpreted as evidence of generalizable value, or does it partly reflect a validation-procedure artifact?

Where this section's rigor is not consistently extended is to the review's own earlier discussion of AI-driven multi-modal phenotyping (section 3.2) and AI/big-data-driven personalization (section 6.4). Both sections describe machine-learning systems integrating imaging, genomic, proteomic, and clinical data to generate treatment recommendations, without applying the same leakage, external-validation, or dataset-shift vocabulary introduced later in section 5.6. This is not a factual error — the review does not misrepresent what these systems can currently do — but it is an internal inconsistency in how much epistemic scrutiny different sections of the same paper apply to structurally similar claims, a predictive model's reported performance, depending on where in the paper the claim appears.

The review's own methodology as a narrative synthesis.

Assessed against the paper's own organizing logic, the classification scheme is coherent as a taxonomy but underdetermined as an evaluative tool. It tells a reader what a biomarker is (sample type, detection dimension, molecular modality) without providing a parallel structure for what evidentiary stage that biomarker occupies. Section 3.5, "Summary of Biomarker Suitability Across Clinical Contexts," is the paper's one explicit attempt at this kind of synthesis, matching biomarker classes to clinical contexts by relative strength, and it is a genuinely useful section precisely because it is the only place in the paper where the authors commit to a comparative judgment rather than a descriptive inventory. Most of the paper does not do this.

5.5 My Perspective

The clearest assumption this paper disrupted for me was the implicit equivalence I had been carrying between "this biomarker is reported in the literature" and "this biomarker is validated." Working through where each of this review's central examples actually sits — MGMT methylation at routine implementation, ctDNA fragmentomics at an earlier clinical-validation stage, microbiome markers not yet past correlative discovery — made clear that "reported" spans an enormous evidentiary range, and that a review organized primarily by molecular modality rather than by evidentiary stage will not surface that range on its own. I had underestimated how much interpretive work a reader has to do to reconstruct that pipeline independently.

The distinction that became sharper for me specifically is clinical validity versus clinical utility. Before this close reading I would have treated "this marker predicts outcome" and "this marker should change treatment" as closer together than they actually are. The MGMT and fragmentomics comparison above is what made the gap concrete: both are clinically valid in the sense the review presents them, but only one currently changes what a clinician does.

The methodological issue that gained importance for me is proportionality in limitation reporting. Before this analysis I would have treated a limitation mentioned anywhere in a paper as functionally equivalent to a

limitation given its own paragraph. The metabolomics example changed that. A limitation's placement and length in the text is itself evidence about how seriously the authors are asking the reader to weigh it, whether or not that was the authors' intent.

6. Limitations

The authors directly state that validation remains inconsistent across laboratories, that companion-diagnostic regulatory pathways lag behind therapeutic development timelines, that multi-omics assays remain too costly for routine deployment outside well-resourced centers, and that equitable access to biomarker-guided care is not assured under current cost and infrastructure constraints. They are explicit that inter-laboratory discordance in variant calling is an active problem rather than a resolved one, which is one of the considerations underlying the ESMO Precision Medicine Working Group's recommendation for complementary cfDNA and tissue-based assessment in relevant clinical contexts, rather than a single technical solution. These are limitations the authors themselves acknowledge, and they acknowledge them accurately.

Separately from what the authors state outright, the structure of the review introduces limitations of its own, discussed in 5.2 and 5.4: the absence of consistent evidence-stage annotation, the disproportionate brevity given to some significant limitations relative to the claims they qualify, and an internal inconsistency in how rigorously different sections interrogate similar categories of claim across biological and computational domains. None of these three limitations is acknowledged by the authors, and none is resolved simply by the fact that the underlying translational barriers are discussed elsewhere in the paper — a reader would still need to reconstruct the evidentiary hierarchy independently.

7. Future Directions

The authors' own stated future directions include adaptive trial designs organized around biomarker status, cost-reduction strategies for liquid-biopsy platforms to improve access, federated learning approaches to enable multi-institutional model training without centralizing patient data, and closer integration of organoid, single-cell, and spatial techniques to better characterize tumor heterogeneity.

Independent of what the authors propose, the evidentiary gaps identified above point toward a narrower set of priorities. First, prospective interventional evidence specifically testing whether earlier detection via fragmentomic MRD assays changes outcomes when acted upon — the cited evidence establishes earlier detection, not that acting on it earlier helps, and these are not the same claim. Second, standardized, field-wide reporting of analytical validity metrics at the level of individual assay platforms rather than biomarker classes, since two assays measuring the same underlying biomarker can differ substantially in analytical performance. Third, extension of the leakage-auditing standard the review itself proposes in section 5.6 to the AI-driven multi-modal phenotyping systems it discusses earlier, ideally before those systems are positioned for prospective clinical evaluation rather than after.

8. Lessons Learned

Discovery and validation are not stages of the same activity so much as different activities entirely, requiring different study designs, different statistical standards, and in most cases different research teams. The volume of literature the review surveys under a single heading obscures how much of it belongs to the discovery phase specifically, generating candidate associations rather than validated clinical tools.

Biological signal and clinical utility are evaluable by almost entirely separate criteria, and a biomarker can satisfy the first extensively while never reaching the second. This review's breadth made that distinction

concrete in a way that reading about a single biomarker class would not have, because the paper's own examples span the full range from established (MGMT) to speculative (AI multi-modal phenotyping) without marking the difference explicitly.

Reproducibility concerns are not specific to any one modality. The same underlying failure mode — an evaluation procedure that inadvertently uses information it should not have access to — shows up as inter-laboratory discordance in variant calling, as batch effects in low-biomass microbiome sequencing, and as data leakage in ML-based biomarker models. Recognizing this as one problem with several surface forms, rather than several unrelated problems, is a more useful way to think about validation rigor going forward.

9. Potential Research Extensions

Each extension follows the same structure: the question, why it matters, what data or method could address it, and what specific limitation identified above it responds to.

1. Does earlier ctDNA-based MRD detection, acted upon with treatment escalation, improve recurrence-free or overall survival compared to standard-interval imaging surveillance?

Why it matters: this is the interventional test that clinical validity alone does not provide; earlier detection is not inherently beneficial without demonstrated benefit to earlier action. Data/methodology: a prospective randomized clinical trial in which eligible MRD-positive patients are assigned to biomarker-guided treatment escalation or standard surveillance. Limitation addressed: the clinical validity / clinical utility gap identified in 5.4.

2. How consistent are analytical performance metrics for ctDNA fragmentomic assays across independent platforms and laboratories?

Why it matters: the review does not report platform-level analytical validity, and cross-platform inconsistency would materially affect how much confidence any single reported result deserves. Data/methodology: a structured comparison using shared reference materials, analogous to the NIST Genome in a Bottle mixtures the review cites in a different context, run across multiple fragmentomic assay platforms. Limitation addressed: absence of quantified analytical validity in the review's presentation.

3. Does applying a leakage-audit framework, of the kind the review proposes in section 5.6, to published AI-driven multi-modal phenotyping studies change their reported performance?

Why it matters: this would test directly whether the internal inconsistency identified in 5.4 reflects a real vulnerability or a difference without a practical consequence. Data/methodology: a systematic reanalysis of a sample of published multi-modal phenotyping studies against a Model-Information-Sheet-style checklist. Limitation addressed: the internal-consistency gap identified in 5.4.

4. Does a structured evidence-stage annotation, applied systematically across a biomarker review of this scope, change how readers calibrate confidence in individual claims compared to the current narrative format?

Why it matters: this tests directly whether the presentation-level limitation identified in 5.2 has a measurable effect on reader interpretation, rather than being a stylistic preference. Data/methodology: a comparative reader study presenting the same biomarker content in narrative versus annotated format. Limitation addressed: the absence of consistent evidence-level tagging identified in 5.2.

10. Questions I Still Have

At what point does a biomarker's evidentiary profile — biological plausibility plus repeated correlative association, without prospective interventional evidence — become strong enough to justify incorporation into clinical decision-making rather than remaining a research-only finding, and who should set that threshold?

Does fragmentomic MRD detection add predictive information beyond the clinical and pathological variables already used for post-surgical surveillance, or does its apparent value partly reflect that comparison not having been rigorously made?

When tissue and liquid biopsy results disagree, as the review indicates happens often enough to motivate ESMO's dual-validation recommendation, is the disagreement primarily a sampling artifact of tumor heterogeneity, an analytical limitation of one or both platforms, or some irreducible combination of both — and would the answer differ by cancer type?

If the same reproducibility concerns the review raises about ML-based biomarker discovery were applied retrospectively to the AI-driven multi-modal phenotyping literature it cites approvingly, how much of that literature's reported performance would hold up?

11. How This Paper Influences My Future Projects

Working through this paper's evidentiary gaps in detail changed how I intend to read biomarker literature going forward: not primarily by asking what a paper reports, but by explicitly locating each reported claim on the discovery-to-implementation pipeline before deciding how much weight it should carry. That is a habit I did not have in a disciplined form before this analysis, and applying it here, rather than only reading about it, is what made the difference.

It has also given me a concrete methodological target for future work: building a small evidence-stage annotation scheme, of the kind sketched in section 4.2 and proposed as a research extension in section 9, and testing it against a narrower slice of the biomarker literature than this entire review covers — most plausibly a single modality, such as ctDNA-based MRD detection, where enough primary literature exists to make the exercise genuinely informative rather than speculative.

12. Conclusion

This review's genuine contribution is the scope of the map it draws across sample types, detection dimensions, and molecular modalities — it establishes, credibly, that the biomarker landscape is far broader than any single-modality account would suggest, and that most of the field's difficulty lies past discovery rather than within it. What it does not establish is a way to tell, from the text alone, which of its examples are ready for clinical decision-making and which remain research findings awaiting validation; that distinction has to be reconstructed by the reader, as attempted here for a representative subset of the paper's examples.

The most consequential translational bottleneck the paper surfaces, whether or not it labels it this way, is not any single technical limitation but the absence of a shared evidentiary standard for comparing biomarkers across modalities — the same word, “biomarker,” covers MGMT methylation at routine clinical use and an AI phenotyping system still at the discovery stage. The most important methodological lesson is closely related: reported association, however well cited, is not self-certifying, and a biomarker's placement on the discovery-to-implementation pipeline has to be checked deliberately rather than inferred from how confidently a claim is written.

13. References

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